Name: _____

WORKSHEET

Combination calculations

1 Write the following in the form $\frac{n(n-1)(n-2)\ldots(n-r+1)}{r!}$

a ${}^{7}C_{4}$

b ${}^{6}C_{1}$

c ${}^{4}C_{2}$

d ${}^{9}C_{8}$

e ${}^{16}C_{7}$

f ${}^{10}C_{7}$

2 Calculate the value of the combinations in question **1**.

a

b

c

d

e

f

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3 Calculate the value of the following combinations.

a ${}^{11}C_4$

b ${}^{7}C_3$

c ${}^{12}C_9$

d ${}^{19}C_3$

e ${}^{8}C_4$

f ${}^{13}C_7$

g ${}^{9}C_0$

h ${}^{10}C_{10}$

i ${}^{28}C_1$

j ${}^{14}C_8$

k ${}^{3}C_2$

l ${}^{8}C_6$

4 Explain the difference between:

a ${}^{8}C_2$ and ${}^{10}C_{2.}$

b ${}^{5}C_0$ and ${}^{5}C_5$.

c ${}^{6}C_4$ and ${}^{6}P_4$.

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5 Write the following in the form ${}^{n}C_{r}$.

a The number of ways a committee of 5 people can be made from a group of 17 people.

b The number of ways a selection of 3 chocolates can be made from a group of 15 chocolates.

c The number of ways a group of 12 people can be made from a bus containing 42 people.

d The number of ways a team of 7 people can be made from 18 applicants.

e The number of different selections of 4 cards can be made from a standard deck of 52 cards.

f The number of different pairs of socks that can be formed from a drawer of 29 different socks.

6 Calculate the value of the problems found in question **5**.

a

b

c

d

e

f

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Answers

1 a $\dfrac{7 \times 6 \times 5 \times 4}{1 \times 2 \times 3 \times 4}$

b $\dfrac{6}{1}$

c $\dfrac{4 \times 3}{1 \times 2}$

d $\dfrac{9 \times 8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2}{1 \times 2 \times 3 \times 4 \times 5 \times 6 \times 7 \times 8}$

e $\dfrac{16 \times 15 \times 14 \times 13 \times 12 \times 11 \times 10}{1 \times 2 \times 3 \times 4 \times 5 \times 6 \times 7}$

f $\dfrac{10 \times 9 \times 8 \times 7 \times 6 \times 5 \times 4}{1 \times 2 \times 3 \times 4 \times 5 \times 6 \times 7}$

2 a 35

b 6

c 6

d 9

e 11 440

f 120

3 a 330

b 35

c 220

d 969

e 70

f 1716

g 1

h 1

i 28

j 3003

k 3

l 28

4 a ${}^{8}C_{2}$ is the number of ways two objects can be selected from a group of 8 (28 ways) and ${}^{10}C_{2}$ is the number of ways two objects can be selected from a group of 10 (45 ways).

b ${}^{5}C_{0}$ is the number of ways zero objects can be selected from a group of 5 (1 way), and ${}^{5}C_{5}$ is the number of ways two objects can be selected from a group of 5 (1 way).

c ${}^{6}C_{4}$ is the number of ways four objects can be selected from a group of 6 (15 ways) and ${}^{6}P_{4}$ is the number of ways four objects can be arranged into an order from a group of 6 (360 ways).

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5 **a** ${}^{17}C_5$

b ${}^{15}C_3$

c ${}^{42}C_{12}$

d ${}^{18}C_7$

e ${}^{52}C_4$

f ${}^{29}C_2$

6 **a** 6188

b 455

c 11 058 116 888

d 31 824

e 270 725

f 406

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